On the other hand, the halogens react with water to give an acidic solution. For example, chlorine forms hydrochloric acid and hypo chlorous acid. It is responsible for the antibacterial action of chlorine water, and for its use as bleach.

$$Cl_2 + H_2O \rightarrow HCl + HOCl$$

Fluorine reacts differently to liberate oxygen from water.

$$2F_2 + 2\ H_2O \rightarrow 4HF + O_2$$

In a similar way, compounds of non-metals react with water to give acidic or alkaline solutions. For example, solutions of carbonates are slightly alkaline.

$$CO_3^{2-} + H_2O \rightarrow HCO_3^{-} + OH^{-}$$

Water is an amphoteric oxide. It has the ability to accept as well as donate protons and hence it can act as an acid or a base. For example, in the reaction with HCl it accepts proton where as in the reaction with weak base ammonia it donates proton.

$$NH_3 + H_2O \rightarrow NH_4^{+} + OH^{-}$$

$$HCl + H_2O \rightarrow H_3O^{+} + Cl^{-}$$

Water dissolves ionic compounds. In addition, it also hydrolyses some covalent compounds.

$$SiCl_4 + 2\ H_2O \rightarrow SiO_2 + 4\ HCl$$

$$P_4O_{10} + 6\ H_2O \rightarrow 4\ H_3PO_4$$

Many salts crystallized from aqueous solutions form hydrated crystals. The water in the hydrated salts may form co-ordinate bond or just present in interstitial positions of crystals.

Examples: $[Cr(H_2O)_6]Cl_3$ – All six water molecules form co-ordinate bond

$BaCl_2.2H_2O$ – Both the water molecules are present in interstitial positions.

$CuSO_4.5H_2O$ – In this compound four water molecules form co-ordinate bonds while the fifth water molecule, present outside the co-ordination, can form intermolecular hydrogen bond with another molecule. $[Cu(H_2O)_4]SO_4.H_2O$

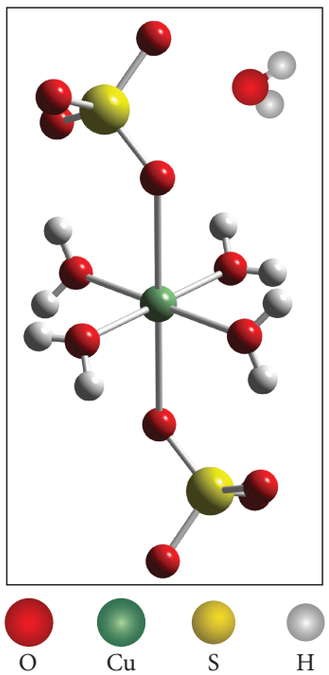


Figure 4.4 structure of copper sulphate pentahydrate

4.5.4 Hard and Soft Water:

Hard water contains high amounts of mineral ions. The most common ions found in hard water are the soluble metal cations such as magnesium & calcium, though iron, aluminium, and manganese may also be found in certain areas. Presence of these metal salts in the form of bicarbonate,

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